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Recommendation Letter

This is to recommend Stanislav Fedotov for participation in Summer School "*Structures in Lie Representation Theory*".

I know Stanislav since September, 2004, he was a student at my algebra classes in 2004/05 and 2005/06 academic years at Moscow State University. His student group was very active, there were brilliant young mathematicians, and Stanislav was among the best.

Since 2006 Fedotov chooses algebra as a speciality, and I am a scientific supervisor of his course and diploma theses. Firstly, I proposed him to study connected components of affine algebraic groups that consist of elements of finite order. It is natural to call such components *periodic*. Stanislav constructed a beautiful theory of algebraic groups with periodic components. His contribution may be considered as a development of classical results due to A.Borel-G.D.Mostow (1955) and R. Steinberg (1968). Recently Stanislav's paper was accepted for publication in Sbornik Math. For these results he has got the finalist diploma of XII Moebius contest.

As a problem for diploma thesis, we decided to take a different subject. In 1994, A.D. King proposed a remarkable construction of moduli spaces for representations of quivers based on Mumford's quotients coming from different linearizations of the trivial line bundle over the representation space. This approach turns out to be very useful. Here the corresponding moduli spaces are quasi-projective varieties. The idea of the diploma thesis is to apply recent generalizations of Mumford's construction leading to non-quasi-projective quotient spaces to representations of quivers. At the moment, Stanislav knows necessary background and technique, and I hope that the project will be successful.

Stanislav is an active participant of our seminars. His talks are well-prepared, clearly presented and contain many interesting facts and results.

In my opinion, Stanislav Fedotov is a gifted and hard-working young mathematician making successfully the first steps in mathematical research. I think that he will fully benefit from attending the summer school. So I strongly support the application Stanislav Fedotov.